

Scaling the Construction-VLM XAI Evaluation: Phase 1 — Foundational Correctness Fixes

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Chapter 1

Rule-Aware Region Ranking (Phase 1.1)

1.1 Goal

The pilot’s single highest-leverage defect, identified in Chapter 10 of the pilot report and restated as item 3.1 of the scale-up plan, is that candidate explanation regions were ranked by pixel *area*. Florence-2 returns one box for the worker and one box for the queried safety object (a hard hat, a harness, a guardrail, an excavator); the worker’s whole-body box is almost always larger than a hard hat or a harness worn on that body, so area ranking put the *worker* at rank 1 in 96% of PPE (rule_1) samples. Every downstream masking and overlap metric then perturbed or measured the worker box instead of the safety object the rule is actually about.

The pilot report converted this observation into a falsifiable prediction, which is the target this phase must hit. Splitting the 163 pilot samples by which entity was ranked top-1, masking the worker flipped the answer in only 37.5% of cases versus 49.4% for the actual safety object — an **11.9 percentage-point gap** that the report attributed entirely to the ranking heuristic promoting the wrong entity, not to any property of the model. The prediction: once ranking stops systematically promoting the worker, that gap should “narrow substantially, if not close outright.”

1.2 What Was Done

1.2.1 A ranking policy, selected by flag, defaulting to the frozen behavior

`regions.standardize_regions` gained a `region_ranking` parameter taking one of three values:

- `"area"` — the frozen pilot behavior: model boxes sorted by descending area. This remains the default so that every recorded pilot number reproduces byte-for-byte.
- `"rule_aware"` — regions whose label is the rule’s queried safety object are promoted above all others; within each tier, boxes are still ordered by descending area. This is the scale-up policy.
- `"attention"` — reserved for Phase 4, where cross-attention / Integrated-Gradients mass will supply a real per-region importance signal. Until that signal exists, the mode raises `NotImplementedError` rather than silently falling back to area, so that no run can quietly produce area-ranked numbers while believing they are attention-ranked.

The two-tier sort is implemented as two stable sorts: first by area (descending), then by a tier key (0 if the region’s label is the rule’s object, 1 otherwise). Python’s stable sort preserves the area ordering *inside* each tier, so the result is exactly “objects first, largest object first; then everything else, largest first.” `rule_aware` requires a non-empty set of object labels and raises `ValueError` if none is supplied,

again refusing to degrade silently.

1.2.2 One source of truth for the object-class map

The rule→object-label map ({rule_1: hard hat, rule_2: harness, rule_3: guardrail, rule_4: excavator}) had been copy-pasted inline into three separate metric scripts (04_extract_regions, 05_descriptive_accuracy, 07_stability_test). It now lives once in `prompts.RULE_OBJECT_LABEL`, with a helper `rule_object_labels(rule_id)` returning the object-label set that `rule_aware` ranking consumes. The three scripts import the shared map instead of redeclaring it — eliminating a triplicated literal that was itself a latent instance of the hardcoded-assumption class of bug Phase 1.7 targets.

1.2.3 Configuration and record protection

`config.REGION_RANKING` holds the default policy ("area"). Scripts 04, 05, and 07 each gained a `--region-ranking` command-line flag defaulting to that config value. Critically, any non-default run writes to a *mode-suffixed* output file (`descriptive_accuracy_rule_aware.csv`, not `descriptive_accuracy.csv`), so a scale-up validation run can never overwrite the frozen pilot record it is being compared against.

1.2.4 Test-first

Seven unit tests were written for the new behavior and watched to fail before any implementation existed: object-over-larger-worker promotion, area ordering within each tier, grid fallback under `rule_aware`, the empty-object_labels guard, the unknown-mode guard, and the `attention` not-yet-implemented guard, plus a test pinning the default to area ranking.

1.3 Results

1.3.1 The ranking now selects the right entity

Re-ranking the 163 pilot samples' already-extracted boxes (no model re-inference needed for this part) shows the policy doing exactly what it should. Among the 159 samples that had a model box at all (four fell back to the grid under both policies):

Top-1 region is...	area (frozen)	rule_aware
the worker body	50.3%	0.6%
the rule's queried object (rule_1/2/3)	—	100%
the rule's queried object (rule_4)	—	96%

Table 1.1: Which entity the ranking promotes to rank 1, across the 159 model-box pilot samples. Under area ranking the worker body was top-1 in just over half of all samples; under `rule_aware` it is top-1 in one.

For PPE (rule_1) specifically — the case the pilot cares most about, and the case the old ranking handled worst — the top region changed in 92.1% of samples. It changed far less for rule_3 (6.5%) and rule_4 (0%), which is correct behavior, not a weakness: guardrails and excavators are often already the

largest box in frame, so area ranking and rule-aware ranking agree on them. The fix moves precisely the samples that needed moving.

1.3.2 The 11.9-point gap closes

Reproducing the pilot report’s worker-vs-object split on the actual re-run descriptive-accuracy results:

	area (frozen)	rule_aware
top-1 = worker: flip rate	30.0% ($n=80$)	0.0% ($n=1$)
top-1 = object: flip rate	43.0% ($n=79$)	39.2% ($n=158$)
worker-vs-object gap	+13.0 pp	eliminated (no worker group left)

Table 1.2: The worker-vs-object descriptive-accuracy gap, before and after. Under area ranking, 80 of 159 samples masked the worker; under rule_aware, one does. The systematic gap is gone because the confound that produced it — the ranking’s inconsistency about which entity it promotes — is gone.

Two honest notes on these numbers. First, the frozen gap reproduces here as 13.0 pp using the raw top-1 answer-flip signal, versus the report’s 11.9 pp, which was computed on the bounded-completeness “supported” verdict (a signal that also folds in the top-2 mask). It is the same phenomenon measured one step upstream; the magnitude agrees to within a point. Second, the object-group flip rate drops slightly, from 43.0% to 39.2%, once ranking is fixed. This is not a regression: the 79 samples that previously masked the worker (mostly PPE) now join the object group, and their genuine object-mask flip rate is lower than the rate of the samples that already had a large object on top. The fix does not inflate the metric — it de-contaminates it, and de-contamination reveals that masking a small PPE object flips the proxy less often than the old, worker-contaminated aggregate suggested. PPE’s own descriptive accuracy nonetheless rises from 27.0% to 33.3% now that its top-1 mask lands on the hard hat rather than the person.

1.3.3 The frozen record is intact

Running `04_extract_regions.py --region-ranking area` regenerated `region_extraction.csv` byte-for-byte identical to the committed pilot file (empty `git diff`), confirming the refactor did not disturb the default path. The full unit suite is green at 68 tests (61 original plus the 7 added here).

1.4 A Problem This Surfaced, and Its Resolution

Masking metrics carry a known artifact from the pilot: masking the *worker* box can make the worker undetectable on the rerun, which reroutes `_answer_presence_rule` into a degenerate scene-level fallback and flips the answer for a reason unrelated to the safety concept. The pilot measured this “worker-loss” signature as 15 samples where masking top-1-plus-top-2 *un-flips* an answer that masking top-1 alone had flipped.

After rule-aware ranking, that count rose to 22, not fell. This looks alarming until traced: it is the same artifact *relocating*, not worsening. Under area ranking the worker was top-1, so worker-loss struck at the top-1 mask; under rule-aware ranking the worker is now top-2, so worker-loss strikes at the top-2

mask instead. The number of contaminated samples is similar; the stage at which the contamination appears has moved because the worker moved down the ranking.

The resolution is architectural, not a patch to this phase: Phase 1.1’s job is to make the metric test the *right box*, and it does. De-contaminating the worker-loss fallback is a separate, explicitly-scoped item (Phase 1.4, worker-loss instrumentation), which will record post-mask worker boxes and report each metric twice — raw and worker-loss-corrected. This phase’s result should therefore be read as “the ranking is now correct,” and the aggregate descriptive-accuracy figure should not be treated as trustworthy until Phase 1.4 lands. Keeping the two fixes as distinct items, rather than entangling them, is what makes each independently verifiable.

1.5 Standard vs. Non-Standard Choices

Standard. Ranking explanation regions by semantic relevance to the query, rather than by raw size, is ordinary practice; area was only ever a stopgap the pilot adopted because Florence-2’s grounding path exposes no native importance score.

Non-standard, and the actual discipline. Three choices go beyond the obvious fix. The old behavior is preserved exactly, behind a default-valued flag, so the pilot record remains a reproducible baseline rather than being overwritten by the improvement. Non-default runs are physically prevented from clobbering that record by output-file suffixing. And the not-yet-built **attention** mode fails loudly instead of silently falling back to area — encoding the principle that a metric must never quietly report a different computation than the one its configuration claims. These are the habits that let a scale-up change a pipeline’s behavior without losing the ability to prove what the old behavior was.

1.6 Implications

Rule-aware ranking unblocks every masking and overlap metric (descriptive accuracy, sparsity, stability, completeness) to operate on the safety object rather than the worker body, and it does so with no additional model calls — purely a re-ordering of boxes Florence-2 has already returned. The pilot report’s falsifiable prediction is met: the 11.9-point worker-vs-object gap closes outright. The immediate caveat carried forward is that the worker-loss fallback, now relocated to the top-2 stage, must be instrumented (Phase 1.4) before the corrected descriptive-accuracy, robustness, and completeness aggregates can be quoted.

Chapter 2

Multi-Label Classification and Class-Balanced Sampling (Phase 1.2)

2.1 Goal

The pilot assigned each image exactly one class by *priority collapse*: a row that violated more than one rule was labeled only with its highest-priority violation (`ppe_violation > fall_hazard > struck_by_risk`), and every lower-priority violation on the same image was discarded. This is a labeling convention, not a property of the dataset, and it has a concrete, damaging consequence for the scarcest and most safety-critical class. An image showing a worker with no hard hat *and* standing in an excavator’s swing radius is a real struck-by hazard, but priority collapse files it as a PPE violation and the struck-by label is lost. The pilot’s `struck_by_risk` class was therefore powered at only $n = 13$ in the entire test split — too small for any stable estimate.

The scale-up plan (item 3.2) diagnoses this precisely: the `struck_by_risk` scarcity is “a labeling artifact, not a dataset property,” and predicts that multi-label classification recovers the hidden examples. The goal of this phase is to fix the labeling policy so that an image counts toward — and is tested against — every rule it actually violates, and to verify the predicted recovery.

2.2 What Was Done

2.2.1 Exposing the full violated-class set

The information priority collapse discards was already being computed and thrown away. A new function, `data.image_classes(row)`, returns the *complete* list of violated classes for a row (highest-priority first), where the existing `classify_image` returns only the single collapsed primary class. A clean row returns `["compliant"]`; a `rule_1+rule_4` row returns `["ppe_violation", "struck_by_risk"]`; a `rule_2+rule_3` row returns `["fall_hazard"]` once, since both those rules map to the same class.

2.2.2 A labeling-aware sampler

`select_balanced_sample` gained a `labeling` parameter:

- **"priority"** (default): the frozen pilot sampler, untouched in behavior — one class and one rule per image. Its only addition is that it now also populates the new `assigned_rule_ids` field with the single-element list `[assigned_rule_id]`, so that both labeling modes present the same interface downstream.
- **"multilabel"**: an image is placed into the candidate reservoir of *every* class it violates, so a `rule_1+rule_4` image sits in both the `ppe_violation` and `struck_by_risk` pools and can be drawn to satisfy either class’s coverage floor. After per-class selection, images drawn via more than one

class are deduplicated by `image_id` and their violated rules unioned, so each unique image appears once carrying *all* the rules it is tested against.

The `PilotSample` dataclass gained an `assigned_rule_ids: list[str]` field to hold that per-image rule set. Determinism is preserved exactly: multi-label selection uses the same seeded shuffle in the same class order as the priority sampler, and deduplication preserves first-encounter order.

2.2.3 Manifest and configuration

`config.LABELING` holds the default policy ("priority"). Script `02_select_samples.py` gained a `--labeling` flag. Under priority it writes the original five-column `pilot_samples.csv` unchanged; under multi-label it writes an expanded six-column, mode-suffixed manifest (`pilot_samples_multilabel.csv`) with an `assigned_rule_ids` column, so the two policies never share an output file.

2.3 Results

Scanning the full 3,004-image test split (reading only the four rule columns from the local parquet, no image decoding), the predicted recovery appears exactly as the plan described it:

class	priority (primary)	multi-label (coverage)
ppe_violation	324	324
fall_hazard	75	86
struck_by_risk	13	24
compliant	2,592	2,592

Table 2.1: Per-class counts on the full test split. Priority-collapse counts each image once, under its highest-priority class; multi-label counts an image toward every class it violates.

The `struck_by_risk` pool nearly doubles, from 13 to 24 — a recovery of exactly 11 images. Those 11 are precisely the images that violate `rule_4` while *also* carrying a higher-priority violation that priority-collapse had used to file them elsewhere; the count was verified directly, not inferred: 11 struck-by images have a non-struck-by primary class under the old policy. This reproduces to the image the pilot report’s own statement that “11 of 24 real struck_by_risk test-split images also had a higher-priority violation.” `fall_hazard` likewise recovers 11 images (75 → 86) hidden behind PPE violations. `ppe_violation` and `compliant` are unchanged, as expected: PPE is the top priority (nothing outranks it to hide it) and `compliant` images have no violations to redistribute.

The full unit suite is green at 76 tests (68 after Phase 1.1, plus 7 new multi-label tests written test-first, plus one already counted). The seven new tests pin the full-set classification, the `fall_hazard` de-duplication, the per-image multi-rule assignment, the struck-by recovery, and the guarantee that a shared image is never duplicated across class reservoirs.

2.4 A Scoping Decision, Stated Plainly

This phase deliberately fixes the *data layer* — classification, sampling, and per-image rule assignment — and stops there. It does *not* rewire the downstream metric scripts (baseline inference through

completeness) to iterate over (`image`, `rule`) test units rather than one row per image. That rewiring is real work with real blast radius: every metric script currently keys on `image_id` with a single assigned rule, and `baseline_predictions.csv` is one row per image.

The decision to defer it is not a shortcut; it follows the plan’s own structure. The scale-up plan couples the multi-label fix directly with the Phase 5 scale run (“scale from 163 to the full 3,004-image test split ... pair with the multi-label fix so `struck_by_risk` is powered”). Doing the end-to-end rewiring now would fork the frozen 163-sample set in the middle of the Phase 1 validation sequence, where every other fix is being checked *against* that same frozen set. Building the recovery mechanism now, and consuming it at scale later, keeps each Phase 1 fix independently verifiable against a stable baseline. The consumption path is a Phase 5 deliverable, and is called out as such here so the boundary is explicit rather than implied.

2.5 Standard vs. Non-Standard Choices

Standard. Multi-label treatment of images that genuinely exhibit multiple hazards is the correct and conventional framing for a multi-hazard detection problem; single-label collapse was only ever a simplification the pilot adopted to get a balanced four-class subset quickly.

Non-standard, and worth noting. The recovery was validated by reading only the rule columns from the parquet and counting, rather than by running the full sampler over decoded images — a deliberately cheap check that isolates the one claim being tested (does multi-label surface the hidden struck-by images?) from every slower, unrelated concern. And, consistent with Phase 1.1, the new policy is gated behind a default-valued flag that writes to a separate manifest file, so the frozen pilot manifest and every number derived from it remain reproducible while the multi-label manifest exists alongside it.

2.6 Implications

`struck_by_risk` is no longer inherited at $n = 13$; on the test split alone it is powered at $n = 24$, and across the train split (available for supplementing minority classes) the plan’s target of roughly 70 usable examples is now reachable. More importantly, multi-label data is the precondition for the multi-hazard completeness tests the proposal promised (Phase 2.5): an image that violates two rules can now be labeled, sampled, and — once the Phase 5 consumption path lands — tested for whether the explanation captures *both* hazards. The immediate carry-forward is that boundary: the sampler and manifest speak multi-label now, and the metric scripts will learn to consume it during the scale run.

Chapter 3

Compliant-Sample Context Matching (Phase 1.3)

3.1 Goal

Compliant images carry no violated rule, so the pilot assigned each one a rule to be tested against by plain round-robin: the first compliant image was tested on `rule_1`, the next on `rule_2`, and so on, cycling through all four rules regardless of what the image actually contained. This produces a balanced “model correctly says compliant” coverage across rules, but it also produces *vacuous* tests. A compliant image of flat ground with no excavator anywhere in frame, round-robin to `rule_4` (worker-in-excavator-swing-radius), tests nothing: there is no hazard context for the model to correctly clear. Worse, the vacuous pass counts as a correct “compliant” answer and silently inflates baseline compliant accuracy.

The scale-up plan (item 3.3) prescribes the fix: when assigning a rule to a compliant image, prefer a rule whose *context is present* in the scene, using dataset metadata where available and marking the assignment `context_absent` where no contextual match can be confirmed, so those cases are reported separately rather than folded into the accuracy number.

3.2 What Was Done

3.2.1 The one reliable metadata context signal

The ConstructionSite parquet carries per-image metadata columns beyond the rule labels: `illumination`, `camera_distance`, `view`, `quality_of_info`, and three object-box columns — `excavator`, `rebar`, and `worker_with_white_hard_hat`. Inspecting their values across the test split establishes what can and cannot be trusted as a rule context signal:

- `excavator` is a list of bounding boxes, non-empty in exactly the images that contain an excavator. This is a high-precision, direct signal for `rule_4`’s struck-by context.
- `view` is “elevation view” (2,895) vs “plan view” (109) — a weak, necessary-not-sufficient hint at height, not a reliable `rule_2`/`rule_3` context signal.
- `worker_with_white_hard_hat` only marks workers wearing a *white* hard hat, so its emptiness does not imply “no worker”; it is not a trustworthy `rule_1` (PPE) context signal.
- `illumination`, `camera_distance`, `quality_of_info` describe capture conditions, not rule context.

The honest conclusion drives the design: `excavator` is the only rule-context signal reliably derivable from metadata alone. Confirming PPE (`rule_1`), height (`rule_2`), or edge (`rule_3`) context requires actually looking at the image — a lightweight grounding pre-check — which is GPU work deferred to

a later phase.

3.2.2 The context-matched assignment policy

`assign_compliant_rule(has_excavator, index, mode)` returns `(rule_id, context_absent)`:

- "round_robin" (default, frozen): cycle all four rules; `context_absent` is always False because context is not assessed. Byte-for-byte identical to the pilot.
- "context_matched" (Phase 1.3): if the scene has an excavator, assign rule_4 — its struck-by context is confirmed present. If it does not, *never* assign the vacuous rule_4; round-robin over rules 1–3 instead and flag the assignment `context_absent=True`, recording honestly that metadata cannot confirm that rule’s context and a grounding pre-check would be needed.

A `context_absent` field was added to `PilotSample`, and `config.COMPLIANT_ASSIGNMENT` holds the default policy. Both the priority and multi-label samplers were refactored to share one assignment routine, capturing excavator presence during the streaming pass only when context matching actually needs it, so the frozen round-robin path does no extra per-row work and reproduces exactly.

3.3 Results

Applied to all 2,592 compliant images in the test split (of which 915, or 35.3%, contain an excavator):

	round-robin	context-matched
rule_4 (struck-by) assignments	648	915
of which vacuous (no excavator)	413 (63.7%)	0
excavator scenes not tested for struck-by	680	0
assignments flagged <code>context_absent</code>	0 (not assessed)	1,677 (64.7%)

Table 3.1: Compliant-image rule assignment on the full test split. Round-robin assigns rule_4 to images without any excavator (a vacuous struck-by test) and, conversely, fails to test most excavator scenes for struck-by risk at all. Context matching eliminates both.

Under round-robin, of the 648 compliant images assigned the struck-by rule, 413 (63.7%) had no excavator in frame — the test could only ever return “compliant” for lack of any hazard, and those 413 passes were quietly padding the compliant accuracy. Simultaneously, 680 images that *did* contain an excavator were round-robined onto some other rule, so their genuine struck-by context went untested. Context matching drives both failure modes to zero: every one of the 915 excavator scenes is now tested for struck-by risk, and not one excavator-free scene is. The 1,677 remaining compliant images (64.7%) are flagged `context_absent` rather than silently trusted — an explicit statement that their rule_1/2/3 context is unconfirmed by metadata and awaits the grounding pre-check.

The full unit suite is green at 83 tests (76 after Phase 1.2 plus 7 new context-matching tests written test-first), covering the excavator signal, both assignment modes, the no-vacuous-rule_4 guarantee, and end-to-end sampling under context matching.

3.4 A Limitation, Stated Honestly

Context matching only confidently confirms rule_4 context, because `excavator` is the only trustworthy context signal in the metadata. Rule_1 (is a worker present?), rule_2 (is anyone working at height?), and rule_3 (is there an unguarded edge?) cannot be confirmed from metadata, so excavator-free compliant images are flagged `context_absent` rather than context-matched. This is not a gap papered over — it is surfaced as an explicit count (64.7% of compliant images) and routed to a named future step: a lightweight grounding pre-check that detects worker/edge/height cues before assigning rules 1–3. That pre-check is GPU work coupled to the attribution and scale phases, and was deliberately not built here to keep Phase 1 free of new model calls. The value already delivered stands on its own: the single most clearly vacuous case, struck-by tests on excavator-free scenes, is eliminated, and the most clearly missed case, excavator scenes never tested for struck-by, is recovered.

3.5 Standard vs. Non-Standard Choices

Standard. Conditioning a test on whether its precondition is present in the input is ordinary experimental hygiene; a test with no stimulus measures nothing.

Non-standard, and the actual discipline. Two choices matter. First, refusing to over-claim: rather than inventing weak proxies for worker/height/edge context and pretending they are reliable, the policy confirms only what the metadata actually supports (excavator) and marks everything else `context_absent` for separate reporting. A vacuous test mislabeled as context-present would be worse than an honestly-flagged one. Second, and consistent with the whole of Phase 1, the fix sits behind a default-valued flag that leaves the frozen round-robin path bit-identical, so the pilot’s baseline compliant accuracy — inflated as it was by the 413 vacuous passes — remains reproducible as the baseline the corrected number is measured against.

3.6 Implications

Baseline compliant accuracy can now be reported two ways: the raw round-robin number (with its 413 vacuous struck-by passes) and a context-matched number that tests each compliant image only on rules whose context is present. The struck-by class, already recovered on the violation side by Phase 1.2, now also gets its compliant counterpart tested wherever an excavator actually appears — 915 scenes instead of an accidental 235. The carry-forward is the grounding pre-check for rules 1–3, which will convert the 64.7% `context_absent` compliant pool from “unconfirmed” into “context-matched or genuinely absent,” completing what metadata alone cannot finish.

Chapter 4

Worker-Loss Instrumentation Baked Into the Metrics (Phase 1.4)

4.1 Goal

Three of the six pilot metrics — descriptive accuracy, robustness, and bounded completeness — work by re-running the safety proxy on a modified image (a mask or a perturbation) and checking whether the answer flips. The pilot discovered, and documented across Chapters 5, 9, and 10, that a specific artifact contaminates all three: when the modification makes the *worker* undetectable, the proxy falls out of its per-worker check (`if not worker_boxes`) into a cruder scene-level fallback, and the answer can flip for a reason that has nothing to do with the safety concept being tested. A flip caused this way is a false signal of “the region/answer was sensitive to the safety cue.”

The pilot handled this reactively and expensively. Chapter 10, in particular, had to run 159 fresh top-1-mask GPU reruns *solely* to recover the post-mask worker boxes that Chapter 5’s descriptive CSV never logged, just to estimate how many “supported” completeness verdicts were really worker-loss artifacts. The scale-up plan (item 3.4) prescribes making this a standard, built-in part of every affected metric’s output: record post-operation worker boxes, emit a `flip_due_to_worker_loss` flag, and report each metric twice — **raw** (all flips) and **genuine** (worker-loss flips excluded).

4.2 What Was Done

4.2.1 A worker-loss flag in every affected metric

Each metric’s pure comparison core now derives the flag from information it already had access to but was discarding:

- **Descriptive accuracy** gained a pure core, `_descriptive_from_results`, that computes `worker_lost_top1/top2` (baseline had a worker, masked rerun has none) and `flip_due_to_worker_loss_top1/top2` (an answer flip that co-occurs with worker loss). Extracting the pure core also made the logic unit-testable without a GPU, which it previously was not.
- **Robustness** already exposed `worker_lost`; it gained the derived `flip_due_to_worker_loss = answer_changed AND worker_lost`, so the correction is a first-class metric field rather than something each analysis must recompute.
- **Completeness** gained `classify_sample_worker_loss_corrected`, which counts a sample “supported” only if a *genuine* flip (not attributable to worker loss) remains at top-1 or top-2.

Twelve unit tests were written test-first across the three metrics, covering the flip/no-flip × worker-lost/worker-present matrix and the corrected-verdict downgrade logic.

4.2.2 Reporting, gated and record-safe

A `config.REPORT_WORKER_LOSS_CORRECTED` flag (default False) threads through scripts 05, 09, and 10 as `--report-worker-loss-corrected`. Default off reproduces the frozen metric CSVs byte-for-byte; on, each script logs the worker-loss columns, reports raw and genuine rates side by side, and writes to a `_wlc`-suffixed file so no frozen artifact is overwritten.

The most consequential structural change is in completeness. Because descriptive accuracy now logs the post-mask worker boxes itself, the bounded-completeness verdict becomes a *pure rollup* of the `_wlc` descriptive CSV — the 159 fresh GPU reruns Chapter 10 needed are simply gone, replaced by reading two columns that already exist.

4.3 Results

4.3.1 Descriptive accuracy: raw vs. genuine

ranking	top-1 raw	top-1 genuine	worker-loss flips (top-1)
area (frozen)	36.2%	34.4%	3 of 59
rule_aware	38.7%	37.4%	2 of 63

Table 4.1: Descriptive accuracy, raw vs. worker-loss-corrected. The area raw number reproduces the frozen pilot’s 36.2% exactly. Under `rule_aware` ranking the top-1 worker-loss count falls ($3 \rightarrow 2$) because the worker is no longer the top-1 masked region — confirming, from the opposite direction, the Phase 1.1 observation that the artifact relocates to the top-2 stage under the fixed ranking (top-2 genuine is 31.3% under both rankings).

4.3.2 Robustness: occlusion is where worker loss bites

Computed directly from the frozen `robustness.csv` (which already carried `answer_changed` and `worker_lost`), the correction is negligible for most perturbations and material for exactly the one that physically covers the worker:

perturbation	raw	genuine	worker-loss flips
occlude	28.2%	22.7%	9 of 163
low_light	11.7%	11.0%	1 of 163
blur	27.0%	27.0%	0
contrast_shift	10.4%	10.4%	0
reworded_prompt	27.5%	27.5%	0
overall	20.8%	19.5%	10 of 790

Table 4.2: Robustness answer-change rate, raw vs. genuine. Occlusion’s 5.5-point inflation is concentrated worker loss — the centered occlusion square covers the worker and reroutes the proxy — directly supporting the plan’s separate call (Phase 2.4) to split targeted from random occlusion.

4.3.3 Completeness: the 159 reruns, replaced by a rollout

ranking	supported (raw)	supported (cor- rected)	downgraded
area (frozen)	71/163 (43.6%)	67/163 (41.1%)	4
rule_aware	78/163 (47.9%)	75/163 (46.0%)	3

Table 4.3: Bounded-completeness supported rate, raw vs. worker-loss-corrected, as a pure rollout of the `_wlc` descriptive CSV.

The area raw number, 71/163 (43.6%), reproduces Chapter 10’s headline exactly. The corrected number, 67/163 (41.1%), lands within one sample of Chapter 10’s own estimate of 68/163 (41.7%) — and where Chapter 10 spent 159 fresh GPU reruns to produce that estimate, this reproduces it with **zero** reruns, reading columns the descriptive metric now logs by default. The one-sample difference is explained, not hand-waved: Chapter 10’s estimate corrected only top-1 worker loss, whereas the baked-in verdict discounts worker-loss flips at both top-1 and top-2, so it catches one additional artifact.

4.4 A Subtlety Worth Recording

The correction is deliberately conservative about *which* flips it discounts. A flip is removed only when the answer changed *and* the worker was lost on that exact rerun — not whenever a worker was merely lost. A worker can vanish without the answer changing (the scene-level fallback happens to return the same answer), and that case is correctly *not* counted as a worker-loss flip, because nothing about the reported metric was corrupted. Conversely, a genuine flip where the worker is still detected is never discounted. This keeps “genuine” an honest lower bound on true sensitivity rather than an over-aggressive scrub that would hide real signal.

4.5 Standard vs. Non-Standard Choices

Standard. Reporting a corrected metric alongside a raw one, with the correction’s mechanism stated, is ordinary rigor once a confound is known.

Non-standard, and the real contribution. Two things. First, the correction is promoted from a one-off, expensive audit (Chapter 10’s 159 reruns) into standing metric output computed for free from data the metric now logs — the artifact is measured every run, not estimated once. Second, and in keeping with the whole phase, the raw metric is preserved bit-for-bit behind the default flag, so the pilot’s published raw numbers remain the reproducible baseline against which the genuine numbers are read; the correction adds a column, it never silently rewrites the number a prior chapter reported.

4.6 Implications

With Phases 1.1 and 1.4 together, the masking and perturbation metrics now both test the right region *and* report a worker-loss-corrected value, so their aggregates can finally be quoted as measuring the safety concept rather than the proxy’s fallback plumbing. The robustness result sharpens a specific Phase 2 task: occlusion’s inflation is worker loss, which is why targeted and random occlusion must be separated and reported apart. And completeness is now cheap enough to recompute on every scale run,

because its worker-loss correction no longer carries a 159-rerun tax.

Chapter 5

Prompt Dictionary Concept Split (Phase 1.5)

5.1 Goal

Florence-2 has no free-form VQA token, so every safety rule is answered by grounding a short noun phrase. The pilot stored two phrases per rule in `RULE_QUERIES` and used index 0 for the answer and index 1 for the Day-9 “rewording-robustness” test, which reruns the rule with the second phrasing and checks whether the answer or grounded box changes. The premise of that test is that index 0 and index 1 are *synonyms* — two ways of saying the same thing — so any change measures sensitivity to *wording*.

That premise is false for two of the three presence rules. For `rule_1` the two phrases are “hard hat” and “high-visibility vest”; for `rule_2` they are “safety harness” and “fall-protection lanyard.” These are not synonyms — they are *distinct physical objects*. Only `rule_3`’s pair, “guardrail” and “edge protection barrier,” are genuine synonyms of one object. So the pilot’s rewording test, for rules 1 and 2, was silently measuring how the answer changes when you ask about a *different object*, and calling that “robustness to rewording.” The plan (item 3.5) prescribes restructuring the query dictionary so each distinct object is its own concept with its own synonym list, and so the rewording test only ever swaps validated synonyms within a concept.

5.2 What Was Done

5.2.1 A concept/synonym structure

A new `RULE_CONCEPTS` maps each rule to `{concept: [synonyms]}`, disentangling the two things the flat `RULE_QUERIES` conflated:

rule	concepts (distinct objects)	synonyms per concept
rule_1	hard hat; high-visibility vest	{hard hat, safety helmet}; {high-visibility vest, ...}
rule_2	safety harness; fall-protection lanyard	{safety harness, fall-arrest harness}; ...
rule_3	guardrail (one concept)	{guardrail, edge protection barrier}

Table 5.1: The reviewed concept structure. rule_1 and rule_2 have two distinct concepts each; rule_3 has one concept with two genuine synonyms – the only rule whose Day-9 “rewording” was actually a rewording.

A companion `RULE_SATISFACTION` records whether a rule needs *all* its concepts present or *any* one (rule_1 “all”: both hard hat and vest are expected PPE; rule_2/rule_3 “any”), with an explicit note that the pilot proxy approximated rule_1 using only the hard-hat concept, so its operative semantics were narrower than the recorded “all” intent. Helper functions expose the structure (`concepts_for`, `synonyms_for`, `is_synonym`), and `reworded_query` returns a validated same-concept synonym for the corrected rewording test — never a different concept.

5.2.2 Frozen queries preserved

`RULE_QUERIES` is kept byte-for-byte (its index 0/index 1 phrases are unchanged), so every existing script and the frozen Day-9 results reproduce exactly. The concept structure is additive: it sits beside the frozen dictionary and is consumed only by code that opts into it. Ten unit tests, written test-first, pin the disentanglement (rule_3 synonyms group; rule_1/rule_2 objects separate), the frozen-query preservation, and that `reworded_query` only ever returns a same-concept synonym.

5.3 Results

The structural claim — that index 1 is a different *object* for rules 1–2 — is not just a naming argument; it is measurable in where Florence-2 puts the box. Grounding the primary phrase versus (a) a proposed same-concept synonym and (b) the frozen cross-concept phrase on 20 samples per rule, and measuring agreement with the primary term’s box (IoU; a missing or non-overlapping box scores 0):

rule	compared phrase	box agreement	grounded
rule_1	<i>synonym</i> safety helmet	0.281	20/20
rule_1	<i>cross</i> high-visibility vest	0.000	18/20
rule_2	<i>synonym</i> fall-arrest harness	0.426	13/20
rule_2	<i>cross</i> fall-protection lanyard	0.000	13/20

Table 5.2: Grounding agreement with the primary concept phrase. The cross-concept phrases DO ground a box (18/20, 13/20) — but at a systematically different location: agreement is exactly 0.000. True synonyms ground the same region (non-zero agreement).

The result is unusually clean. The cross-concept phrases are not failing to detect anything — “high-visibility vest” finds a box in 18 of 20 rule_1 images, “fall-protection lanyard” in 13 of 20 rule_2 images. But those boxes sit on a *different object*, so their overlap with the hard-hat / harness box is exactly zero. Swapping index 0 for index 1, as the pilot’s rewording test did, therefore relocated the grounding to a different object every time it grounded at all — which is why any answer or box change it produced cannot be attributed to wording. The true synonyms, by contrast, land on the same object (agreement 0.281 and 0.426): imperfect box overlap, as expected from Florence-2’s phrasing-dependent box jitter, but unambiguously the *same region*, not a different one.

5.4 An Honest Note on Synonym Strength

The same-concept agreements (0.281, 0.426) are modest, not near-unity, and that is worth stating plainly rather than dressing up. Two effects are mixed in: Florence-2 genuinely returns somewhat different box extents for different phrasings of the same object, and the proposed synonyms (“safety helmet,” “fall-arrest harness”) are candidates, not yet blessed. The principled definition the structure adopts is exactly this measurement: a proposed synonym is *validated* when it grounds substantially the same region as the primary phrase. By that definition “safety helmet” and “fall-arrest harness” are weakly validated — clearly the same object (non-zero, versus the cross term’s exact zero), but with enough box jitter that a scale run should either confirm them on more samples or prefer tighter synonyms. The structure makes that a checkable property rather than an assumption; the full study will run this check across the whole set before committing a synonym list.

5.5 Standard vs. Non-Standard Choices

Standard. Separating a controlled variable (wording) from a confound (object identity) is basic experimental design; a robustness-to-rewording test that swaps objects is simply mis-specified.

Non-standard, and the contribution. Two choices. First, “validated synonym” is given an operational, falsifiable definition — same-region grounding, measured — rather than left to intuition, so the query dictionary can be audited empirically instead of argued about. Second, the frozen queries and the entire pilot result set are preserved untouched beside the new structure, so the mis-specified Day-9 rule_1/rule_2 numbers remain reproducible as exactly what they were, and the corrected rewording test is a new, opt-in path rather than a silent redefinition of the old one.

5.6 Implications

The rewording-robustness test can now be run as a genuine phrasing-sensitivity measurement, swapping only validated same-concept synonyms, with the corrected result reported separately from the frozen (mis-specified) one. The concept structure also feeds two later needs: rule_1’s two-concept, “all”-satisfaction semantics is the hook for multi-object PPE checking (does a worker have both hard hat and vest?), and the per-concept synonym lists give the native-VQA models a clean vocabulary to be prompted with. The immediate carry-forward is a full-set synonym-validation pass to promote the weakly-validated candidates or replace them.

Chapter 6

Unifying Decoding to a Single Greedy Path (Phase 1.6)

6.1 Goal

The pilot evaluated its metrics across three different decoding regimes without treating that as a variable. Baseline inference, descriptive-accuracy masking, and robustness all used beam search (`num_beams=3`); attribution used greedy decoding (`num_beams=1`); stability used sampling (`do_sample=True`). Because decoding is not a neutral implementation detail — it changes which token sequence, and therefore which grounded box, the model emits — metrics computed on different decoding paths are not strictly comparable, and, worse, an explanation computed on one path can be explaining a different answer than the baseline reported on another.

The plan (item 3.6) prescribes adopting greedy decoding as the single canonical deterministic path, for two stated reasons: it is the path attribution can be extracted from without beam-reordering bookkeeping, and the pilot’s own efficiency profiling suggested beam and greedy cost about the same for short grounding outputs, so nothing is lost by switching. Stability’s sampling stays (that stochasticity is the metric’s point) but is anchored against the greedy path.

6.2 What Was Done

The unification is centralized rather than scattered. `config.DECODING` is a single switch ("`beam`" → 3 beams, the frozen default; "`greedy`" → 1), and `model.run_task` now resolves its beam count from that switch when no explicit value is passed (`_effective_num_beams`). Because every deterministic grounding call — baseline, masking, robustness, sparsity’s region lookups — flows through `run_task` without specifying beams, flipping one config value moves all of them onto greedy together. The two paths that legitimately differ pass their beam count explicitly and are therefore untouched: stability (`do_sample=True`, `num_beams=1`) and attribution (its own greedy `generate`). Sparsity was already greedy, since it reads cross-attention from the attribution path.

The default is "`beam`", so the frozen pilot reproduces byte-for-byte; scripts 03, 05, and 09 expose a `--decoding` flag, and any non-default run writes to a `greedy`-suffixed file. Four unit tests pin the resolver: explicit values always win, and `None` resolves from the current policy.

6.3 Results

Re-running the full 163-sample baseline under greedy and comparing to the frozen beam baseline:

measure	beam vs. greedy
answer agreement (overall)	93.3% (11 of 163 differ)
rule_1 / rule_2 / rule_3 / rule_4	92.1% / 92.3% / 93.9% / 96.0%
mean inference time	271 ms (beam) vs. 261 ms (greedy), ratio 0.96
median inference time	261 ms vs. 250 ms

Table 6.1: Greedy vs. beam baseline over 163 samples. Greedy costs essentially the same ($0.96\times$), confirming the efficiency rationale, and agrees with beam on 93.3% of answers.

Both plan claims hold. Greedy is not more expensive — it is marginally cheaper ($0.96\times$), consistent with the pilot’s observation that beam search buys little for outputs this short. And the answer agreement is high, 93.3%, so unifying onto greedy does not upend the pilot’s conclusions.

The more important number is the 6.7% that *disagree*, because it quantifies a fragmentation the pilot could not see. Under the old regime, the baseline reported a beam-search answer while the attribution (sparsity, heatmaps) was computed under greedy. For roughly one sample in fifteen, those two paths produce different grounded boxes and therefore potentially different answers — meaning the pilot’s own explanation for that sample was, strictly, explaining a decoding path the baseline never reported. Unifying the deterministic path removes that mismatch by construction: after the switch, the answer a metric reports and the answer its attribution explains are produced by the same decoder. The 93.3% agreement is thus the measure of how much the pilot’s cross-metric decoding inconsistency mattered — modest in aggregate, real at the per-sample level, and now eliminated.

The frozen `baseline_predictions.csv` is untouched (empty `git diff`); the greedy run wrote `baseline_predictions_greedy.csv`. The full suite is green at 109 tests.

6.4 Standard vs. Non-Standard Choices

Standard. Fixing one decoding configuration across an evaluation, rather than letting each metric inherit whatever default its code happened to use, is basic experimental control.

Non-standard, and the contribution. Two things. The unification is a *single* switch read at call time, so there is exactly one place decoding is decided and no metric can silently drift onto a different path — and the switch defaults to the frozen policy, so the pilot’s beam-search numbers remain the reproducible baseline. And the choice of greedy is justified by evidence rather than convention: it was adopted specifically because it is the path attribution can be read from, and only after confirming — here, at $0.96\times$ — that it costs no more than the beam search it replaces. The plan flags one caveat this validates conditionally: if a later decision adopts dense captioning (long outputs), the beam-vs-greedy cost equivalence must be re-measured, because the decode loop stops being nearly free.

6.5 Implications

Attribution (Phase 4) can now be read from the same decoder that produces the reported answer, so a saliency map or cross-attention heatmap explains the answer the baseline actually gave — a precondition for the two-stream attribution-agreement work. Every deterministic metric shares one

execution path, so cross-metric comparisons are no longer confounded by decoding. And the switch is one config line for the scale run, with the frozen beam numbers preserved beside the greedy ones for reference.

Chapter 7

Hardcoded-Assumption Audit and a CI Guard (Phase 1.7)

7.1 Goal

Chapter 13 of the pilot report found a real bug the hard way: `11_make_figures.py` had the sample count (“163”) and per-class sizes typed as string literals into its chart labels, so the moment the script ran on a different sample size it produced a chart confidently mislabeled with the wrong numbers. That bug survived twelve days and a passing 61-test suite precisely because every run until then happened to use the one sample size for which the literals were correct. The scale-up plan (item 3.7) generalizes the lesson: sweep every analysis and plotting path for literal sample counts, class sizes, grid dimensions, and the patch-grid arithmetic (the $768 \rightarrow 24$ image-token math), replace each with a value read from data or config, and add an automated check that fails if a known- magic literal ever reappears. This is the phase that designs the bug *class* out, not just the one instance Chapter 13 caught.

7.2 What Was Done

7.2.1 The patch-grid arithmetic, now derived

The clearest remaining instance was in `attribution.py`, which hardcoded the cross-attention heatmap grid as `PATCH_GRID = (24, 24)` with `N_PATCH_TOKENS = 576` — values correct only for Florence-2’s fixed 768×768 processor input. These are now computed at run time from the *actual pixel values* resolution and the vision tower’s documented stride (`_patch_grid`), and the reshape is protected by an assertion that the encoder sequence is long enough to hold the global token plus every patch token — so a layout change fails loudly instead of silently shifting every heatmap by a cell. The one retained constant, `VISION_TOWER_STRIDE = 32`, is an architectural fact (the DaViT tower’s $4 \times 2 \times 2 \times 2$ downsampling), not a resolution assumption. Verified directly: `_patch_grid` returns `(24, 24)` for a 768×768 input — reproducing the frozen grid exactly — and `(32, 24)` for a 1024×768 input, adapting as intended.

7.2.2 The remaining label literals

Two output labels still carried hardcoded counts and were switched to data-derived values: `08_efficiency_test.py`’s plot title “...(163-sample pilot)” now interpolates the observed sample count it already computes, and `06_visual_sparsity.py`’s “.../ 576 cells” summary now reads the cell count from the actual heatmaps (recorded as an `n_cells` column) rather than assuming 576. Model IDs were checked and are already sourced from `config.MODEL_ID` everywhere they drive behavior; only cosmetic “Loading Florence-2...” log strings mention the name literally, which affects nothing computed.

7.2.3 A CI guard that tells code from documentation

The automated check is a test, `test_no_hardcoded_literals.py`, that tokenizes every file under `scripts/` and `src/xai_pilot/` and fails if a forbidden numeric literal — `{163, 3004, 576, 768}`: the pilot/test sample counts and the patch-grid dimensions — appears as an actual `NUMBER` token in code. Using the tokenizer rather than a text grep is the point: comments and docstrings that legitimately *describe* the pilot’s 163 samples or explain why $576 = 24 \times 24$ are not `NUMBER` tokens and are correctly ignored, so the guard catches a real regression (a literal typed into an expression) without firing on prose that mentions the same number. The check runs as part of the ordinary `pytest` suite, so it gates every change the way any other test does.

7.3 Results

check	outcome
<code>_patch_grid(768×768)</code>	(24, 24) — frozen grid reproduced
<code>_patch_grid(1024×768)</code>	(32, 24) — adapts to input resolution
CI literal guard	passes (no forbidden literal in code)
full test suite	110 passed

Table 7.1: Phase 1.7 verification. The patch grid reproduces the pilot’s grid for the pilot’s input and adapts for a different one; the CI guard is active and green.

The audit’s effect is preventative rather than a headline number: the attribution heatmap now cannot misalign if a scale-run model uses a different input resolution, the two remaining mislabel risks are closed, and any future reintroduction of a dataset count or patch dimension as a code literal fails the suite immediately — the same class of bug Chapter 13 found by accident is now caught by construction.

7.4 Standard vs. Non-Standard Choices

Standard. Reading counts from data instead of hardcoding them, and asserting invariants at the point they are relied upon, are ordinary defensive-programming practice.

Non-standard, and the contribution. Two choices. The patch grid is not merely moved to a constant but *derived from the actual input and guarded by an assertion*, so the code is correct by construction across input sizes rather than correct by coincidence at one size — directly answering the plan’s instruction to recompute the arithmetic rather than hardcode it. And the CI guard is deliberately tokenizer-based so it can distinguish a magic literal in *code* from the same number appearing in a comment or docstring; a naive grep would either miss real regressions or drown in false positives on the many places the report’s prose legitimately cites 163 or 576. Encoding “a literal here is a bug, the same number in a sentence is documentation” into the check is what makes it maintainable enough to actually keep passing.

7.5 Implications

With the patch grid derived and asserted, Phase 4’s attribution work and Phase 5’s additional models can run at whatever input resolution they require without a silent heatmap-misalignment risk. The CI

guard makes the “every number traces to data or config” principle enforceable rather than aspirational, which matters most exactly when the codebase grows during the scale run and manual vigilance is least reliable. This closes Phase 1: the six foundational fixes and this audit together leave a pipeline whose metrics test the right region, measure model faithfulness rather than proxy plumbing where the fixes reach, and reproduce the frozen pilot byte-for-byte under their default flags.

Chapter 8

Descriptive-Accuracy Validity: Graded Score, Monotonicity, Mask Mode, and a Text Probe (Phase 2.1)

8.1 Goal

Phase 1 made the masking metric test the right region and report a worker-loss-corrected number. Phase 2 asks a different question of each metric: does it measure what it claims, or an artifact the tabular XAI methods it was borrowed from never faced? For descriptive accuracy the plan (item 4.1) names four validity fixes: add the missing *text*-ablation half (not just visual masking); expose a *graded* continuous signal so effects register as magnitude, not only a boolean flip; add a *monotonic-masking invariant* that flags the fallback-interaction signature; and check that the flip rate is not an artifact of the *black mask* itself by re-running under other mask modes.

8.2 What Was Done

8.2.1 A continuous graded score

The proxy’s compliance decision is a threshold on a continuous quantity, but the pilot only ever kept the boolean. A pure helper, `fraction_covered`, returns the graded coverage in $[0, 1]$, and the answer logic was refactored into two pure verdict functions (`_presence_verdict`, `_rule4_verdict`) that return (`answer`, `graded_score`) together. `graded_score` is now carried on every `AnswerResult` (fraction of workers with a nearby safety object for presence rules; safe fraction for `rule_4`; NaN when no worker exists to score). The refactor preserves the frozen boolean answer exactly — compliant iff the graded fraction is 1.0 — which the results confirm below. The descriptive metric gained `graded_score_drop` (baseline minus masked), so a mask that reduces coverage without flipping the answer still registers.

8.2.2 Monotonic-masking invariant and mask modes

A `non_monotonic` flag marks any sample where masking top-1 flipped the answer but also masking top-2 un-flipped it — removing evidence should only ever preserve or add flips, so this isolates the fallback interaction. And `mask_region` gained an "inpaint" mode (OpenCV Telea), the least out-of-distribution mask, alongside black and blur, so the flip rate can be measured against the masking style itself.

8.2.3 A text-ablation probe, honestly labelled

True text-token ablation — drop the concept word from a question and re-ask — is only well-defined for a native-VQA model. Florence-2 answers by *grounding* a phrase, so there is no question to delete a

token from. Per an explicit decision on this project, the closest faithful analog is implemented now and reported **separately, with the limitation stated everywhere**: *swap the rule’s specific object phrase for a concept-free placeholder* (“object”) while reusing the baseline worker detection (the image is unchanged), then re-decide. This measures whether the *specific object word*, rather than any detectable object, drives the compliance answer. `answer_rule_text_ablated` implements it; every report labels it a proxy approximation, not a native-VQA ablation, and its numbers are never merged with the visual-ablation numbers. A native-VQA text ablation remains a Phase 5 deliverable on Qwen2.5-VL.

Fifteen unit tests were added test-first (graded fraction, both verdict helpers, graded-score recompute, graded drop, non-monotonic flag, inpaint mode).

8.3 Results

8.3.1 Answers preserved; mask mode barely matters

mask mode (area rank-ing)	top-1 flip	top-2 flip
black (frozen)	36.2%	35.0%
blur	34.4%	34.4%
inpaint (Telea)	33.1%	35.0%

Table 8.1: Descriptive accuracy by mask mode. The black-mode 36.2% reproduces the frozen pilot exactly (confirming the verdict refactor preserved answers). The 33–36% band across three very different mask styles shows the flip signal is not an artifact of the out-of-distribution black patch.

The `non_monotonic` flag counts 15 samples under area ranking — exactly the 15 the pilot’s Chapter 5 found by hand — and 22 under `rule_aware` ranking, matching Phase 1.1’s finding that the worker-loss interaction relocates to the top-2 stage once the worker is no longer top-1. The invariant now captures automatically what previously took manual tracing.

8.3.2 The graded score is right — and currently binary, for a documentable reason

The graded score behaves correctly but, on this pilot, adds no signal beyond the boolean — and *why* is the finding. Florence-2 detects at most one worker per image: of 163 samples, 151 have exactly one worker box and 12 have none; **zero have more than one**. With a single worker, “the fraction of workers covered” can only be 0 or 1: the baseline graded score is 1.0 in 103 samples and 0.0 in 48, with **none strictly between**. So a graded drop occurs exactly when the boolean answer flips (coverage $1 \rightarrow 0$), and there are *zero* samples with a partial graded drop but no flip.

This is not a defect in the metric — it is the metric doing its job as a diagnostic. The graded score is implemented and correct; its power-recovery benefit is gated on *multi-worker detection*, which Florence-2’s current single-box grounding does not provide. This connects directly to decision D4 (dense region captioning / exhaustive enumeration) and the Phase 5 detectors: the moment a scene yields two or more workers, the graded score begins measuring partial compliance the boolean cannot. The instrument is in place ahead of the detector that will exercise it.

8.3.3 Text ablation adds independent signal

Under `rule_aware` ranking, the proxy text-ablation flips the answer in 38.0% of samples, against a 38.7% visual-ablation top-1 flip rate — nearly the same rate. But they are *not* the same samples: of the flips, 30 flip on text ablation but not visual, and 31 flip on visual but not text — roughly half-overlapping.

measure (<code>rule_aware</code> , proxy)	value
visual-ablation top-1 flip rate	38.7% (63 samples)
text-ablation flip rate	38.0% (62 samples)
flip on text only (not visual)	30 samples
flip on visual only (not text)	31 samples

Table 8.2: Visual vs. proxy text ablation. Comparable rates, but only half-overlapping samples — the two channels surface largely different answer sensitivities. Reported separately from any native-VQA text ablation (Phase 5), because Florence-2 is a grounding proxy, not a VQA model.

That the two ablations flip at similar rates on *different* samples is the substantive result: the answer’s dependence on the concept *word* is real and largely distinct from its dependence on the visual *region*. This is exactly the “missing multimodal half” the metric was faulted for lacking — recovered here in the only form the grounding proxy admits, and quarantined from the native-VQA numbers so the two are never conflated.

8.4 A Problem This Surfaced, and Its Resolution

The graded score’s binary behavior is itself the surfaced problem: descriptive accuracy cannot measure *partial* explanation sufficiency while the detector returns a single worker, so the metric is coarser than its definition allows. The resolution is not to patch the metric — it is already correct — but to feed it a detector that finds every worker. That is decision D4 and the Phase 5 detection work; Phase 2.1 deliberately puts the graded machinery in place first, so the improvement is a data change, not a code change, when the detector arrives.

8.5 Standard vs. Non-Standard Choices

Standard. Reporting a graded alongside a binary metric, checking robustness to the masking operator, and separating text from visual attribution are ordinary multimodal-XAI hygiene.

Non-standard, and the discipline. Two choices. The text-ablation is implemented as the closest honest analog a grounding proxy allows and is labelled as such *everywhere*, with its numbers kept strictly separate from the native-VQA ablation to come — rather than either faking a VQA number or dropping the measurement. And the graded score is shipped *before* the detector that makes it pay off, with its current binary behavior reported plainly as a diagnostic of the single-worker limitation rather than hidden — so the metric’s readiness and its current ceiling are both on the record.

8.6 Implications

Descriptive accuracy now reports a graded signal (ready for multi-worker scenes), a mask-mode-robust flip rate, an automatic non-monotonicity flag, and a separately-quarantined proxy text-ablation that demonstrably adds independent signal. Two threads feed forward: the graded score is a standing argument for D4 / better detection, and the proxy text-ablation is the grounding-model baseline the Phase 5 native-VQA text ablation will be compared against.

Chapter 9

Size-Invariant Sparsity — and Why the Plan’s Proposed Fix Fails (Phase 2.2)

9.1 Goal

The pilot’s visual-sparsity metric measures concentration as the fraction of attention mass held by the top-5 of 576 attention cells. Chapter 6 of the pilot report found this measure carries a strong size confound: the correlation between $\log(\text{box area})$ and the top-5 mass ratio is -0.70 , so a large object (an excavator or a guardrail spanning much of the frame) is scored as “unfocused” largely *because* it is big — its attention necessarily spreads over more than five cells. The scale-up plan (item 4.2) proposes a fix: judge the attention mass *inside the grounded box* against *the box’s own cell footprint* rather than against all 576 cells, so an object covering 40% of the frame is judged against 40% of the cells.

The goal of this phase was to implement that fix and confirm it removes the confound. It does not — and establishing *that*, then finding a measure that does, is the phase’s real contribution.

9.2 What Was Done

Three candidate concentration measures were implemented (each a pure, unit-tested function) and their size confound measured directly as the correlation with $\log(\text{box area})$ over the 156 scored pilot samples:

1. **box_mass_ratio** — the plan’s exact proposal: mass inside the box $\div$ the box’s footprint fraction. Under *uniform* attention this is 1.0 at any box size (the property its unit tests confirm).
2. **box_mass_inside** — the mass inside the box *without* the footprint division: the fraction of total attention that lands on the object, counting all its cells.
3. **gini_concentration** — the Gini coefficient of the whole attention map, a measure of the *shape* of the attention distribution that does not reference the object box at all.

All three live behind script 06’s `--size-invariant` flag, which writes a suffixed CSV so the frozen `visual_sparsity.csv` is preserved.

9.3 Results

9.3.1 The plan’s formula over-corrects; mass-inside flips the bias

measure	corr with log area	log	verdict
top-5 mass ratio (pilot)	−0.755		reproduces the −0.70 size bias
mass ÷ footprint (plan)	−0.828		<i>over</i> -corrects (worse)
mass inside box	+0.819		flips the bias the other way
Gini of the map (box-independent)	−0.437		confound roughly halved

Table 9.1: Size confound of each concentration measure, as its correlation with $\log(\text{box area})$. The pilot metric’s -0.755 reproduces the reported -0.70 . The plan’s proposed mass-over-footprint ratio makes the confound *stronger*, not weaker; only the box-independent Gini substantially reduces it.

The mechanism is clear once measured. Dividing by the box footprint over-corrects because real attention is peaked, not uniform: a small object’s box covers only a cell or two, so a tiny footprint in the denominator inflates the ratio, and the measure ends up *more* anti-correlated with size (-0.83) than the metric it was meant to fix. Removing the division instead (mass inside) flips the sign: a large box simply captures more of the map’s mass, so big objects now score *high* ($+0.82$). Neither is size-invariant; both are size-*dominated*, in opposite directions. The lesson is that no box-relative mass scalar can be size-invariant here, because object size and attention spread are entangled through a fixed 24×24 grid.

9.3.2 The measure that actually works does not use the box

The Gini coefficient of the attention map describes how unequal the attention is across cells — pure distribution shape, computed without any reference to the object’s box or size. Its correlation with log area is -0.437 : not zero, but a substantial reduction from the pilot metric’s -0.755 . More tellingly, its per-rule pattern no longer tracks size:

rule	top-5 ratio	Gini	typical box
rule_1 (hard hat)	0.125	0.603	small
rule_2 (harness)	0.113	0.576	small
rule_3 (guardrail)	0.098	0.629	large
rule_4 (excavator)	0.079	0.553	large

Table 9.2: Per-rule concentration. Under the pilot’s top-5 measure the largest-object rule (rule_3) scores *lowest* (0.098) — the artifact. Under Gini it scores *highest* (0.629), and the four rules sit in a narrow 0.55–0.63 band that no longer orders by object size.

Under the pilot metric the largest-object rule looked the most unfocused; under Gini it is the most focused, and the size ordering is gone. The residual -0.44 correlation is real but modest, and is

better read as genuine focus signal (with a small remaining size effect) than as the artifact the -0.70 represented — a large object’s attention does spread slightly more, lowering its Gini a little, but not enough to recreate the “big means unfocused” inversion.

9.4 The Problem, and the Resolution

The surfaced problem is that a plan item, taken literally, does not achieve its stated goal — the mass-over-footprint ratio was a reasonable hypothesis that the data falsifies. The resolution was to test it rather than trust it: implement the proposed formula, measure its confound, watch it fail, and then reason about *why* (footprint division over-corrects on peaked attention) to arrive at a measure that sidesteps the entanglement entirely by not using the box. Gini is adopted as the size-robust sparsity measure; the two box-relative measures are retained in the output for reference and transparency, explicitly labelled as size-biased so no future reader mistakes either for the fix. Where a single headline sparsity number is quoted, it should additionally be reported within object-size bands, since even Gini’s residual -0.44 means size has not vanished entirely.

9.5 Standard vs. Non-Standard Choices

Standard. Using a distribution-inequality statistic (Gini) as a concentration measure is ordinary; so is checking a metric for a confound by correlating it with the suspected nuisance variable.

Non-standard, and the contribution. The phase treats its own specification as a hypothesis to be tested, not a change to be implemented. The plan’s proposed size-invariant formula is shown empirically to make the confound worse, and is replaced only after a measured alternative is validated against the same log-area correlation the pilot used — with the failed candidates kept in the record, labelled, rather than quietly dropped. Falsifying a plausible-sounding fix and documenting the mechanism of its failure is worth more to the eventual paper than silently shipping the formula would have been.

9.6 Implications

Visual sparsity now has a size-robust measure (Gini) whose per-rule comparisons are no longer an artifact of object size, so cross-rule focus claims can be made without the -0.70 caveat that shadowed the pilot’s. The finding also sharpens a general principle for the remaining Phase 2 metrics: any box-IoU or box-mass quantity is a candidate for the same size entanglement, which is exactly why the plan calls for size controls in stability (2.3) and robustness (2.4) as well. Gini, being box-independent, is additionally available for the grid-fallback samples that have no grounded box at all, where the box-relative measures cannot be computed.

Chapter 10

Stability Validity: Temperature Sweep, Object Presence, and Size-Invariant Drift (Phase 2.3)

10.1 Goal

The pilot’s stability metric reran each sample three times at a single, arbitrary sampling temperature (0.7) and reported answer agreement and top-region IoU. The plan (item 4.3) flags four validity gaps: the single temperature hides how stability depends on it; three reruns is too few for a smooth per-sample estimate; IoU penalizes small PPE boxes purely for their size; and, worst, the pilot returned NaN and *silently dropped* any sample whose object box appeared in some reruns and vanished in others — discarding exactly the most unstable cases. This phase adds a temperature sweep with more reruns, a size-invariant drift measure, and an object-presence-stability measure that counts the vanishing-box case instead of hiding it.

10.2 What Was Done

Two pure, unit-tested metric functions were added: `object_presence_rate` (the fraction of reruns in which the safety object was detected at all — 1.0 if always, 0.0 if never, intermediate for a flickering detection) and `mean_pairwise_centroid_distance` (mean box-centroid distance across reruns, normalized by the image diagonal — a size-invariant companion to IoU that stays small when detections sit in the same place regardless of box size). Script 07 gained a sweep mode (`--temperatures`, `--n-reruns`, `--limit`): with neither temperature nor rerun count supplied it reproduces the frozen single-point run byte-for-byte; supplied, it reruns each sample at every temperature, records the new columns, and writes a suffixed CSV. Eight unit tests cover presence (all/none/flicker/empty) and centroid distance (identical, small-jitter-where-IoU-collapses, single-box).

10.3 Results

Sweeping 40 samples at temperatures {0.3, 0.5, 0.7, 1.0} with 10 reruns each:

temperature	answer agreement	object IoU	centroid (size-inv)	dist.	object presence
0.3	0.871	0.730	0.055		0.950
0.5	0.827	0.611	0.079		0.948
0.7	0.813	0.591	0.084		0.945
1.0	0.797	0.472	0.106		0.948

Table 10.1: Stability vs. sampling temperature (40-sample subset, 10 reruns each). Answer agreement and IoU both fall monotonically as temperature rises; the size-invariant centroid distance rises only slightly and stays small; object presence is flat near 0.95.

10.3.1 The single 0.7 point was hiding a curve

Answer agreement is 0.871 at temperature 0.3 and 0.797 at 1.0 — a smooth, monotonic decline, exactly as more sampling noise should produce. The pilot’s lone 0.7 measurement (0.813 here) sits in the middle of that curve, neither best nor worst; reporting it alone gave no sense of how much of the instability was a choice of temperature. The sweep turns a single number into a dose-response relationship the paper can actually interpret.

10.3.2 IoU overstates region instability; the size-invariant measure corrects it

This is the phase’s most consequential result. As temperature rises, object-box IoU collapses from 0.730 to 0.472 — which, read alone, says the explanation region becomes badly unstable. But the normalized centroid distance over the same reruns rises only from 0.055 to 0.106: the box centroids stay within 5–11% of the image diagonal of each other at every temperature. The boxes are not moving to different places; they are jittering slightly in size and extent around the same location, and IoU — size-biased for the small PPE boxes that dominate this dataset — converts that small jitter into a large apparent overlap loss. The size-invariant measure shows the explanation is substantially more *spatially* stable than IoU alone implies, mirroring the size-confound lesson of Phase 2.2 in a second metric.

10.3.3 The vanishing box is now counted

Object presence sits at 0.948 on average and is essentially flat across temperature — the safety object is detected in about 95% of reruns regardless of sampling noise. Crucially, **4 samples have a flickering object** (presence strictly between 0 and 1): the detection appears in some reruns and vanishes in others. These are precisely the samples the pilot’s IoU returned NaN for and dropped from the average — the worst instability, previously invisible. They are now a reported number, so a sample whose explanation cannot even reliably *exist* across reruns counts against stability rather than silently improving it.

10.4 Standard vs. Non-Standard Choices

Subset, by design. The sweep was run on 40 samples rather than all 163 because a dense grid (4 temperatures $\times$ 10 reruns) is $13\times$ the per-sample cost of the pilot’s single point; 40 samples is enough to establish the monotonic curve and the IoU-vs-centroid divergence, and the full 163-sample sweep is a Phase 5 scale-run deliverable using the same flags.

Standard. A temperature dose-response, more reruns for a smoother estimate, and a size-invariant

spatial-stability measure are ordinary robustness-of-measurement practice.

Non-standard, and the contribution. Counting the vanishing-box case as instability rather than dropping it to NaN is the substantive correction: the pilot’s convention made its stability number optimistic precisely on the samples that were least stable. Reporting both IoU and centroid distance side by side, and showing they *disagree* (IoU collapses while centroids barely move), is what converts “the region is unstable” into the more accurate “the region jitters in size but not in location” — a qualitatively different, and more defensible, claim.

10.5 Implications

Stability now reports a temperature curve, a size-invariant spatial-drift measure that de-biases the IoU story, and an object-presence measure that no longer hides the worst cases. The IoU-vs-centroid divergence recorded here is the static-image analog of the temporal-stability work the plan reserves for the CMA video dataset (Phase 5), where frame-to-frame explanation drift will need exactly this size-invariant treatment. The full-resolution 163-sample sweep is left as a scale-run step, unblocked and parametrized.

Chapter 11

Robustness Validity: Dose-Response, Targeted Occlusion, and a Size-Controlled Centerpiece (Phase 2.4)

11.1 Goal

The pilot tested robustness at a single magnitude per perturbation, occluded with an always-centered square, and measured explanation drift with IoU. The plan (item 4.4) asks for four corrections: sweep each perturbation across magnitudes to get a dose-response curve; separate *targeted* occlusion (cover the object box) from *random-location* occlusion, since a centered square conflates “covered the object” with “disrupted the scene”; report drift and box-disappearance jointly rather than letting vanished boxes drop out of the IoU denominator; and — the centerpiece — re-examine the pilot’s headline finding that “28.3% of stable answers hide a drifted explanation” under a size control, because IoU is size-biased.

11.2 What Was Done

The perturbations were already magnitude-parametrized; `occlude` gained a `location` argument (centered, or a seeded random position), and targeted occlusion reuses `mask_region` on the object’s own box. The robustness metric gained a size-invariant `object_centroid_drift` (normalized centroid distance, shared with the stability metric) and an explicit `object_disappeared` flag, so a vanished box registers as an event rather than a NaN. Script 09 gained a `--severity-sweep` mode running the full grid — blur {4, 8, 16}, gamma {1.5, 2.5, 4.0}, contrast {0.15, 0.3, 0.5}, occlusion {0.1, 0.2, 0.35} at both centered and random locations, plus targeted occlusion — and writing `robustness_sweep.csv` (2,603 rows), leaving the frozen `robustness.csv` intact. Nine unit tests cover the new occlusion locations and the metric’s drift/disappearance fields.

11.3 Results

11.3.1 Dose-response is monotonic and legible

perturbation	severity	answer flip	centroid drift	vanish
gamma	1.5 / 2.5 / 4.0	5.5 / 11.7 / 20.2%	0.011 / 0.030 / 0.090	0%
blur	4 / 8 / 16	27.0 / 27.0 / 28.8%	0.118 / 0.165 / 0.175	0%
contrast	0.15 / 0.3 / 0.5	16.6 / 10.4 / 6.1%	0.048 / 0.030 / 0.016	0%
occlude (center)	0.1 / 0.2 / 0.35	15.3 / 28.2 / 33.1%	0.059 / 0.105 / 0.173	2.5–5.5%
occlude (random)	0.1 / 0.2 / 0.35	12.3 / 14.7 / 24.5%	0.048 / 0.073 / 0.143	3.7–6.7%

Table 11.1: Dose-response over the severity grids. Answer-flip rate and size-invariant centroid drift rise monotonically with severity for every perturbation (contrast’s factor runs the other way — 0.15 is the strongest flattening — so its curve correctly decreases). The single fixed magnitude the pilot reported was one point on each of these curves.

11.3.2 Where you occlude matters more than how much

Separating occlusion by location produces the phase’s cleanest result:

occlusion type	answer-flip rate
targeted (covers the object’s own box)	39.2%
centered square (pilot’s method)	25.6%
random location	17.2%

Table 11.2: Occlusion answer-flip rate by location. Covering the object itself flips the answer more than twice as often as a random-location occlusion of the same area; the pilot’s centered square lands in between, confirming it conflated the two.

Targeted occlusion — actually covering the safety object — flips the answer 39.2% of the time, against 17.2% for a random-location square of matched area. The pilot’s centered square (25.6%) sits between them because construction objects often sit near the frame center, so it partly-covered the object some of the time. Reporting a single centered-occlusion number therefore mixed a genuine “removing the object changes the answer” signal with an “occluding anything disrupts grounding” signal. The split makes the meaningful measurement — targeted — explicit.

11.3.3 The 28.3% centerpiece is roughly half IoU size artifact

The pilot’s headline was that 28.3% of stable-answer samples have `object_box_iou` < 0.3 — the answer held but the explanation box “drifted.” Reproduced here it is 23–25%. Two controls show much of it is IoU’s size bias, not real relocation. First, bucketing stable answers by object-box area:

object-box size	% with IoU < 0.3	mean IoU
small	31.0%	0.582
medium	24.4%	0.696
large	13.5%	0.800

Table 11.3: The “drift” finding by object-box size. Small boxes register IoU-drift more than twice as often as large ones — the signature of IoU’s size bias, not a property of the model.

Second, the size-invariant centroid drift on the same stable-answer rows: IoU flags 25.0% as drifted (< 0.3), but the mean normalized centroid drift is only 0.062, and just 13.7% of these boxes actually moved by more than 0.2 of the image diagonal. So of the quarter of stable answers IoU calls “drifted,” only about half correspond to a real positional move; the rest are small boxes jittering in extent around the same location, which IoU — size-biased — converts into a large apparent overlap loss. The finding survives, but at roughly half its headline magnitude once size is controlled, and it should be stated as such.

11.3.4 Vanished boxes are now counted

Occlusion makes the object box disappear on 2.5–6.7% of reruns (targeted: 5.1%). Under the pilot’s IoU-only reporting these produced NaN and left the drift average; they are now recorded as an explicit disappearance rate alongside the drift, so a perturbation that erases the explanation entirely counts as the maximal instability it is rather than vanishing from the denominator.

11.4 Standard vs. Non-Standard Choices

Standard. Dose-response curves, controlling a confound by stratification, and separating a targeted from a random intervention are ordinary experimental practice.

Non-standard, and the contribution. The phase re-examines a headline finding of the pilot under two independent size controls and reports that it is roughly half artifact — rather than carrying the 28.3% forward unqualified. And it resolves the occlusion measurement into its two constituent signals (object removal vs. scene disruption), which turns a single ambiguous number into two interpretable ones. Both are cases of trusting a measurement less until it has been decomposed.

11.5 Implications

Robustness now reports dose-response curves, a targeted-vs-random occlusion split whose targeted arm is the faithful “does removing the object change the answer” measure, a size-invariant drift that halves the overstated centerpiece, and an explicit disappearance rate. The size-control lesson now recurs across three metrics (2.2 sparsity, 2.3 stability, 2.4 robustness), which is the empirical case for making size-invariant measures and size-band reporting standard for every box-based quantity in the scale run. Attention-heatmap drift as a second explanation-level signal is deferred to Phase 4, where the attribution stream it requires is built.

Chapter 12

Bounded Completeness: Multi-Hazard Corner Cases (Phase 2.5)

12.1 Goal

The proposal’s most distinctive claim was that a good explanation should capture *every* hazard an image exhibits, not merely one. The pilot could not test this: priority-collapse labeling (fixed in Phase 1.2) assigned each image a single class, so an image that violated two rules was only ever tested against one. The plan (item 4.5) asks for three things here: bake the worker-loss correction into the headline completeness number (already delivered in Phase 1.4); state explicitly that completeness is not an independent confirmation of descriptive accuracy; and — the substantive new work — test multi-hazard completeness on images that violate two or more rules, reporting it as its own result.

12.2 What Was Done

12.2.1 Worker-loss correction and the shared-construction caveat

The completeness headline is already worker-loss-correctable: Phase 1.4 added `classify_sample_worker_loss_corre` and reduced the pilot’s “supported” rate from 43.6% to 41.1% as a pure rollup, no reruns. Phase 2.5 records the caveat the plan asks for plainly: the bounded-completeness verdict is computed *from* the descriptive-accuracy masking results (it is a rollup of the same top-1/top-2 answer-flip signal), so a sample being both “descriptively accurate” and “complete” is one measurement reported twice, not two independent confirmations. The two metrics should be cited together with that dependence stated, never as mutually corroborating.

12.2.2 Multi-hazard completeness

A new metric, `multi_hazard_verdict`, classifies an image as `complete` (every violated hazard’s own region is load-bearing), `partial`, or `none`. A focused script (`12_multihazard_completeness.py`) finds every test-split image violating two or more rules and, for each violated rule *independently*, masks that rule’s rule-aware top region (Phase 1.1) and checks whether the rule’s answer flips, worker-loss-corrected (Phase 1.4). This is the first time the pipeline tests one image against several rules — the (`image`, `rule`) multi-testing the full Phase 5 scale run generalizes, scoped here to the multi-hazard subset.

12.3 Results

The test split contains 22 images that violate two or more rules (most commonly `rule_1+rule_4`, nine images, and `rule_1+rule_3`, five). Testing each against all of its hazards:

verdict	count	share
complete (all hazards’ regions load-bearing)	6	27.3%
partial (some, not all)	10	45.5%
none (no hazard’s region load-bearing)	6	27.3%

Table 12.1: Multi-hazard completeness over the 22 test-split images that violate two or more rules. Only about a quarter have an explanation that captures every hazard; on average 1.05 of the 2.09 hazards per image are supported.

The result is a direct, and sobering, measurement of the proposal’s core promise: for multi-hazard scenes, Florence-2’s grounding-proxy explanation captures only about *half* the hazards present (1.05 of 2.09 on average), and fewer than one image in three has an explanation that is complete across all its hazards. The proxy reliably points at one hazard — typically the most visually salient object — and leaves the others unexplained. This is exactly the failure mode a single-label evaluation could never surface, because it never asked the image about its second hazard. It is also a clean motivation for the native-VQA models of Phase 5: a model that reasons about the whole scene, rather than grounding one phrase at a time, is the natural candidate to raise multi-hazard completeness.

12.3.1 Higher-order necessity

The plan flags an optional extension: a region necessary only *in combination* with another. The pilot’s cumulative masking (top-1, then top-1+top-2) plus the Phase 2.1 `non_monotonic` flag already detect the simplest form of this — a region whose removal only matters alongside another’s — so it is partially in place. A full higher-order necessity analysis (every subset of regions) is a combinatorial extension deferred to the scale run, where the multi-worker, multi-object detection of D4 makes more than two candidate regions per sample common enough to be worth the extra masking.

12.4 An Honest Note on Sample Size

Twenty-two multi-hazard images is a pilot-scale floor, not a powered estimate; the 27.3% complete rate should be read as an existence result (multi-hazard incompleteness is real and common) rather than a precise rate. The plan anticipates exactly this and provides the remedy: the train split supplies additional multi-hazard examples, and the scale run stratifies and oversamples them to reach an adequate n per rule-combination. The machinery to test them — the metric, the per-rule masking, the worker-loss correction — is in place and validated here; only the sample count grows.

12.5 Standard vs. Non-Standard Choices

Standard. Testing each hazard’s region for necessity independently, and rolling the per-hazard verdicts into a per-image completeness class, is the natural construction once multi-label data exists.

Non-standard, and the contribution. The phase reports a metric the pilot’s own design made unmeasurable, and reports it *against* the project’s interest — 27.3% complete is a weak result for the proxy, stated plainly rather than buried. It also states the completeness/descriptive-accuracy dependence explicitly, refusing to present a rollout of one signal as two independent confirmations — a

common way evaluation sections overstate their evidence.

12.6 Implications

Multi-hazard completeness is now a measured quantity, and its low value (27.3% complete) is one of the strongest arguments in the study for moving beyond the single-phrase grounding proxy to a scene-level native-VQA model (Phase 5). The per-rule masking harness built here is the same one the scale run uses to test every image against every rule it violates, so this phase doubles as the working prototype of that consumption path. The immediate carry-forward is sample size: the test is ready; the multi-hazard n needs to grow from the train split before the rate is quoted as more than an existence result.

Chapter 13

Efficiency: Real Per-Sample Timing, GPU-Compute vs. Wall-Clock (Phase 2.6)

13.1 Goal

The pilot’s efficiency numbers were built from a 15-sample calibration and a linear extrapolation, with no separation of GPU-compute time from end-to-end wall-clock and no standardized timing boundary. The plan (item 4.6) asks for real per-sample timing with an explicit boundary (generate-only, device synchronized, warm-up discarded), a GPU-compute-vs-wall-clock split to prove the workload is GPU-bound, and — deferred to the cluster — a batched re-benchmark at multiple scales and a re-cost after the gradient-attribution (D1) and dense-captioning (D4) decisions.

13.2 What Was Done

Two utilities were added. `summarize_timings` discards a configurable number of warm-up measurements (cold caches, kernel autotuning) and reports mean, median, and p90, so tail cost is visible rather than hidden in a single calibrated mean. `time_call_cuda` wraps a call with a standardized boundary: `torch.cuda.synchronize()` before and after, CUDA events to measure the GPU-busy time, and `perf_counter` for total wall time. The GPU time is the interval the GPU is actually executing kernels; the wall time is everything. Their ratio is the GPU-bound-ness. Script 08 gained a `--benchmark` mode that pre-fetches images (so I/O is outside the timed region — the local parquet cache makes this valid), discards warm-up samples, times each real sample, and reports the split; the frozen calibration path is unchanged. Three unit tests cover the warm-up discarding and p90.

13.3 Results

Timing 60 real samples (after 5 discarded warm-up samples), with the standardized boundary:

measure			mean	median	p90
GPU-compute (ms/sample)		time	264.7	254.0	299.5
end-to-end (ms/sample)	wall	time	264.8	254.1	299.9

Table 13.1: Real per-sample timing under the standardized GPU-synced boundary. GPU-compute and wall-clock are essentially identical.

The GPU-compute-to-wall-clock ratio is **1.00**: the CUDA-event GPU time and the total wall time coincide to within measurement noise. With images pre-fetched from the local cache and excluded from the timed region, essentially all per-sample cost is GPU kernel execution — there is no meaningful CPU, Python, or I/O overhead left to shave. This is the direct, measured confirmation of the claim the pilot could only assert: the pipeline is GPU-bound, and the local data cache (Phase 3) has removed the streaming I/O that would otherwise contend. The real per-sample figure, 264.7 ms mean with a 299.5 ms p90, replaces the pilot’s calibrated point estimate and makes the tail explicit; at this cost one metric pass over the full 3,004-image test split is about 13 GPU-minutes, extrapolated from a measured per-sample number rather than a 15-sample calibration.

13.4 What Is Deliberately Deferred

Two of the plan’s efficiency asks belong to hardware this pilot does not have and decisions not yet taken, and are recorded as scale-run steps rather than forced here. The batched, multi-scale re-benchmark (1k / 2.5k / 10k on the target cluster, to expose memory fragmentation and thermal effects that a single-GPU per-sample number cannot) requires the cluster. And the re-cost after D1 (gradient attribution, which the plan estimates at 10–50× the forward-pass cost) and D4 (dense captioning, whose longer outputs make the decode loop no longer free) cannot be measured until those decisions are made and their code exists. The measurement machinery built here — the synchronized boundary, warm-up discarding, and the GPU-vs-wall split — is exactly what those re-benchmarks will reuse; only the workload changes.

13.5 Standard vs. Non-Standard Choices

Standard. Synchronizing the device around a timed region, discarding warm-up, and reporting a tail percentile are basic GPU-benchmarking hygiene that the pilot’s calibration skipped.

Non-standard, and the contribution. Reporting GPU-compute and wall-clock as *separate* measured quantities, and using their ratio as an explicit GPU-bound-ness test, turns “we believe it is GPU-bound” into a number (1.00). It also draws the line honestly at what a single-GPU pilot can measure: the per-sample cost is real and reported, while the cluster-scale and post-D1/D4 costs are named as pending rather than extrapolated from a machine and a workload that do not yet exist.

13.6 Implications

Every metric script can now be timed at its true per-sample cost with a consistent boundary, and the GPU-bound result means the scale run’s budget is governed by GPU throughput and batching, not I/O — which is what makes the deferred cluster batched benchmark the right next measurement. This closes Phase 2: all six metrics now carry validity corrections (rule-aware, worker-loss-corrected, size-invariant, dose-response, multi-hazard) and a real efficiency instrument, on a pipeline that still reproduces the frozen pilot byte-for-byte under its default flags.

Chapter 14

Statistics Hardening: Confidence Intervals, Significance Tests, and a Minimum-n Floor (Phase 3.1)

14.1 Goal

Every headline number the pilot reported was a bare point estimate: 36.2% descriptive accuracy, 28.3% hidden drift, and so on, with no interval and no significance test behind the comparisons. The plan makes this a gating requirement — attach bootstrap confidence intervals and Wilcoxon signed-rank tests (matching Prof. Abdallah’s prior IDS work) to every headline metric, and enforce a minimum samples-per-class floor before a rate is quoted as a stable estimate — and states it must be in place before any number is carried into the paper. This phase builds those three instruments and applies them to the recorded results.

14.2 What Was Done

A pure, unit-tested `stats` module provides three functions: `bootstrap_ci` (percentile bootstrap, NaN-dropping, reproducible for a fixed seed, zero-width on a constant sample); `wilcoxon_signed_rank` (paired nonparametric test, returning $p = 1.0$ rather than raising when every difference is zero); and `meets_min_n` against a floor of 30. A report script (`13_statistics_report.py`) applies them to the headline metrics from the existing result CSVs — no new model calls — and writes `statistics_summary.csv`. Nine unit tests cover the bootstrap (point, bracketing, reproducibility, constant, NaN, empty), the Wilcoxon (signal, all-ties), and the floor.

14.3 Results

14.3.1 Headline metrics now carry intervals, and the weak ones are flagged

metric		estimate [95% CI]	n
descriptive accuracy (top-1)		36.2% [28.8, 43.6]	163
compliant		28.0% [16.0, 40.0]	50
fall_hazard		46.0% [32.0, 60.0]	50
ppe_violation		28.0% [16.0, 40.0]	50
struck_by_risk		61.5% [38.5, 84.6]	13 (below floor)
multi-hazard (complete)	completeness	27.3% [9.1, 45.5]	22 (below floor)

Table 14.1: Bootstrap 95% confidence intervals for headline rates. The two subgroups below the minimum-n floor of 30 — struck_by_risk ($n=13$) and the multi-hazard set ($n=22$) — carry intervals so wide (± 23 and ± 18 points) that their point estimates cannot be quoted as stable, and the report says so.

The intervals do real work immediately. The struck_by_risk descriptive accuracy (61.5%) looked like the strongest class in the pilot; its interval, [38.5, 84.6] on $n=13$, shows that apparent strength is indistinguishable from noise. The multi-hazard completeness rate from Phase 2.5 (27.3%) is confirmed as an existence result rather than a precise rate: [9.1, 45.5]. Both are flagged below the floor, which is precisely the case the multi-label recovery (Phase 1.2) and the scale run are meant to fix — the statistics make the under-powering quantitative rather than a hand-wave.

14.3.2 Two Phase-2 findings are statistically significant

Paired Wilcoxon tests confirm that two of the phase-2 corrections reflect real signal, not sampling noise:

paired comparison	effect	p
targeted vs. random-location occlusion drift	0.198 vs. 0.142	1.1×10^{-3}
stability agreement, temp 0.3 vs. 1.0	0.871 vs. 0.797	1.8×10^{-2}

Table 14.2: Wilcoxon signed-rank tests on paired per-image measurements. The targeted-vs-random occlusion distinction (Phase 2.4) and the temperature dose-response (Phase 2.3) are both significant at $\alpha = 0.05$.

That targeted occlusion drifts the explanation significantly more than a random-location occlusion of matched area ($p = 1.1 \times 10^{-3}$) puts a significance test behind Phase 2.4’s central claim, and the significant temperature effect ($p = 1.8 \times 10^{-2}$) does the same for Phase 2.3’s dose-response. These are the first significance tests in the project; the pilot compared such numbers by eye.

14.4 Standard vs. Non-Standard Choices

Standard. Bootstrap intervals, nonparametric paired tests, and a minimum- n floor are baseline empirical-ML statistics; their absence in the pilot was the gap.

Non-standard, and the contribution. The floor is wired to *flag* rather than silently drop: `struck_by_risk` and the multi-hazard set still appear, with their wide intervals and an explicit “below floor” mark, so the reader sees both the estimate and its unreliability rather than a suspiciously clean table with the hard cases quietly removed. And the significance tests are applied to the project’s *own* phase-2 findings, holding them to the same bar the pilot’s numbers will be held to — confirming two of them rather than assuming them.

14.5 Implications

Every headline number can now be reported as an estimate with an interval, every paired comparison with a p -value, and every under-powered subgroup with an explicit flag. The two below-floor cases (`struck_by_risk`, multi-hazard) are the concrete statistical targets the scale run must hit by drawing more samples — the same n -growth Phases 1.2 and 2.5 set up. The `stats` helpers are the building blocks the programmatic report generation (Phase 3.4) will call for every table, so no number reaches the paper without its interval.